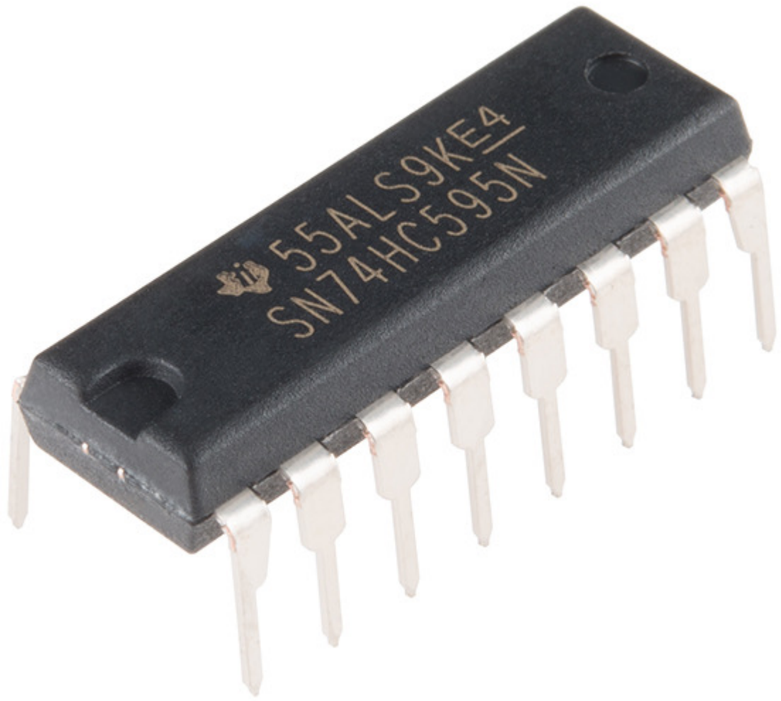
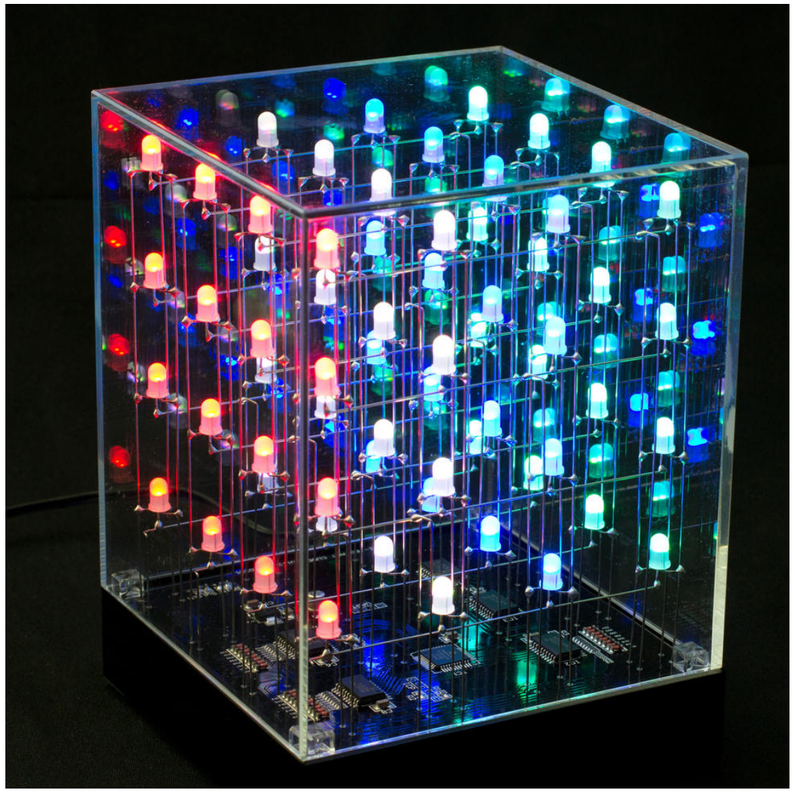
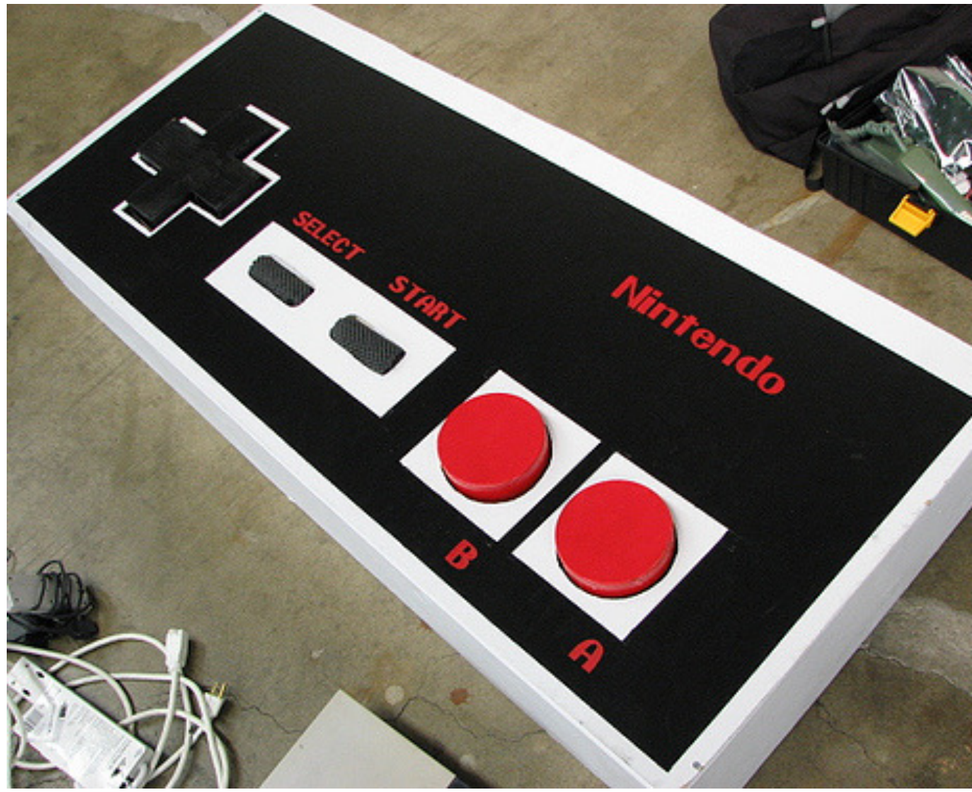
Shift Register

Huaishu Peng | UMD CS | Spring 2026



Shift Register - Why





Shift Register - How

SIPO Vs PISO Shift Registers
Shift registers come in two basic types, either SIPO (Serial-In-Parallel-Out) or PISO (Parallel-In-Serial-Out). The popular SIPO chip is 74HC595, and the PISO chip is 74HC165.

The first type, SIPO, is useful for controlling a large number of outputs, like LEDs. While the latter type, PISO, is good for gathering a large number of inputs, like buttons

https://lastminuteengineers.com/74hc595-shift-register-arduino-tutorial/

Shift Register - How – 74HC595

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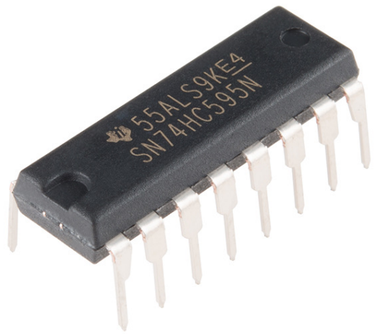

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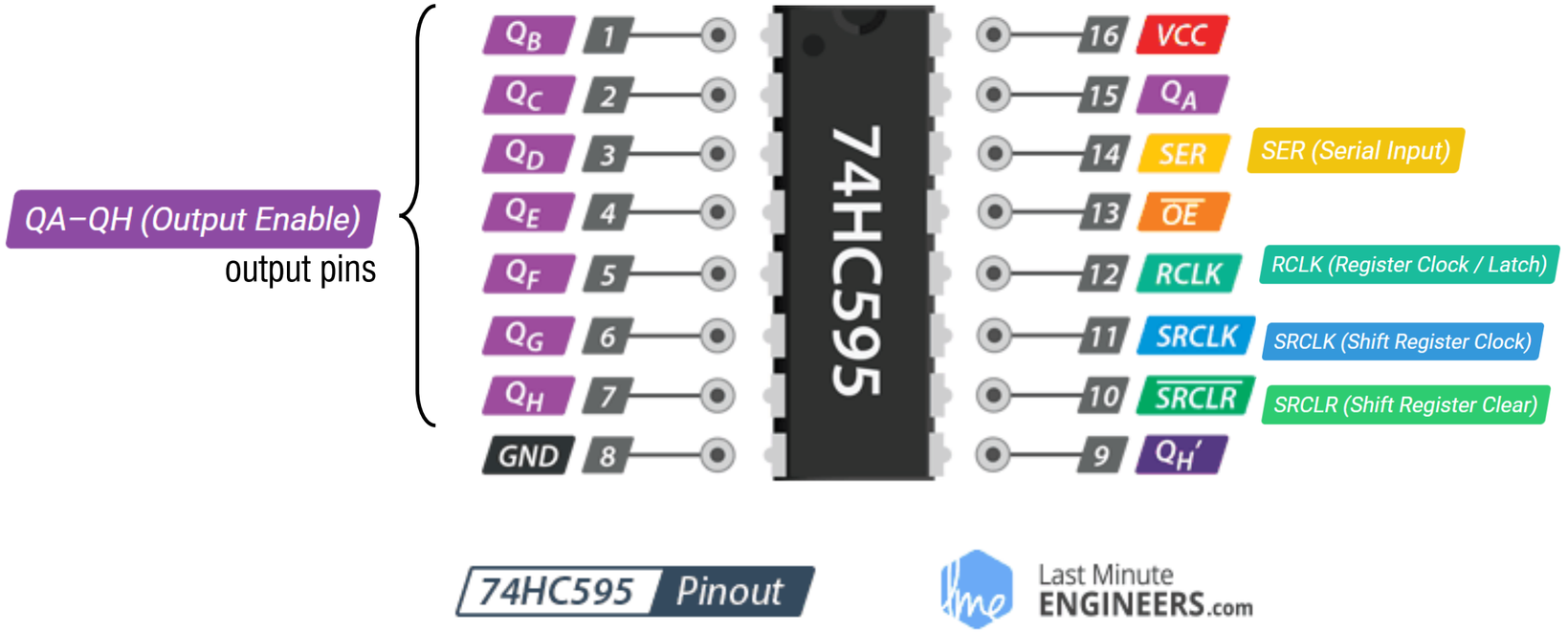
Shift Register - How – 74HC595



to feed data into the shift register a bit at a time

when driven HIGH, the contents of Shift Register are copied into the Storage/Latch Register

the clock for the shift register

allows us to reset the entire Shift Register to 0

74HC595 { 8bit Shift Register, 8bit Storage Register

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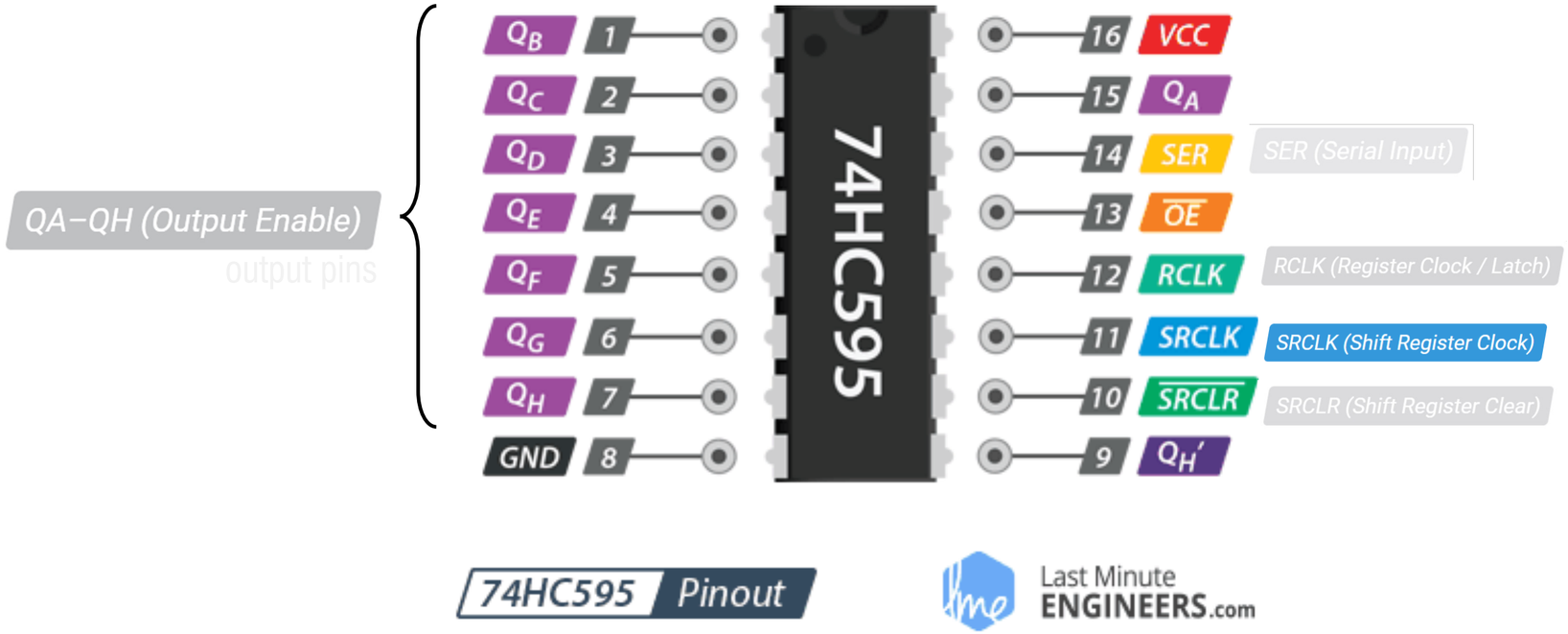
Shift Register - How – 74HC595



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74HC595 { 8bit Shift Register 0b00000001
8bit Storage Register

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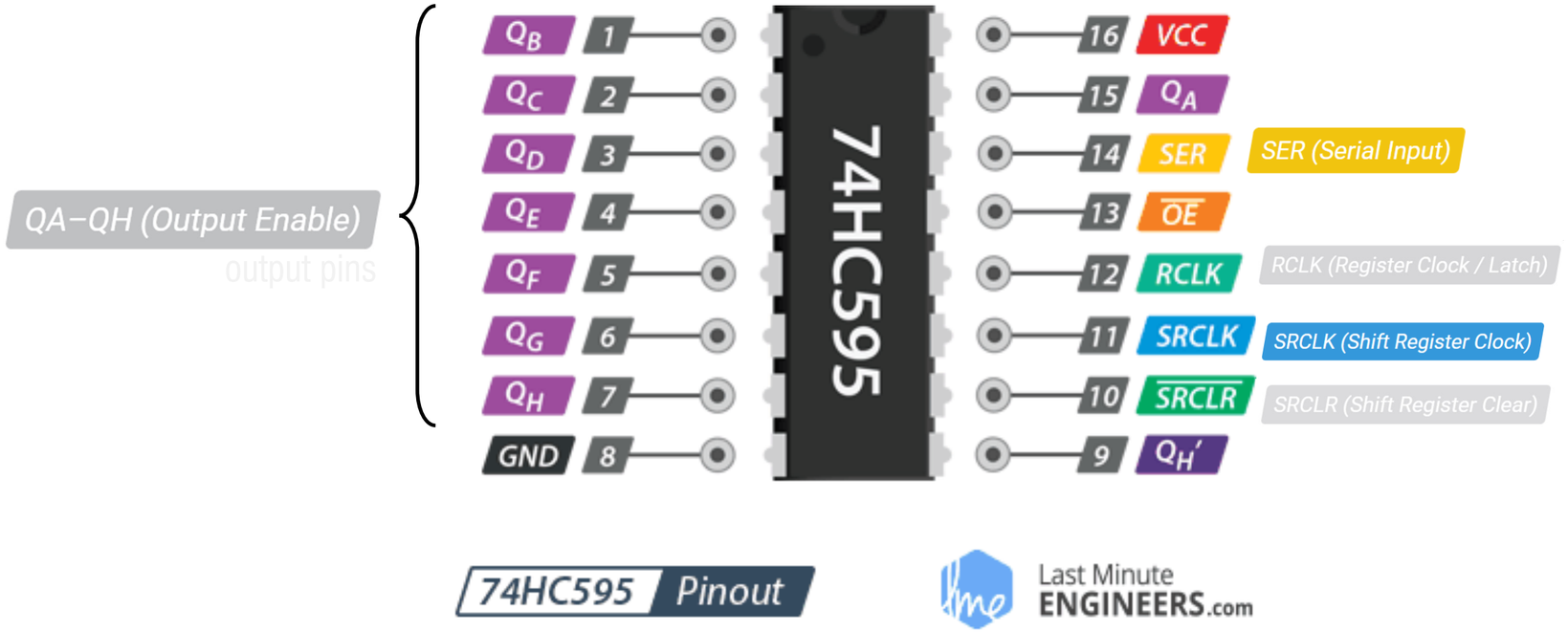
Shift Register - How – 74HC595



to feed data into the shift register a bit at a time (0)

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74HC595 { 8bit Shift Register 0b00000010
8bit Storage Register

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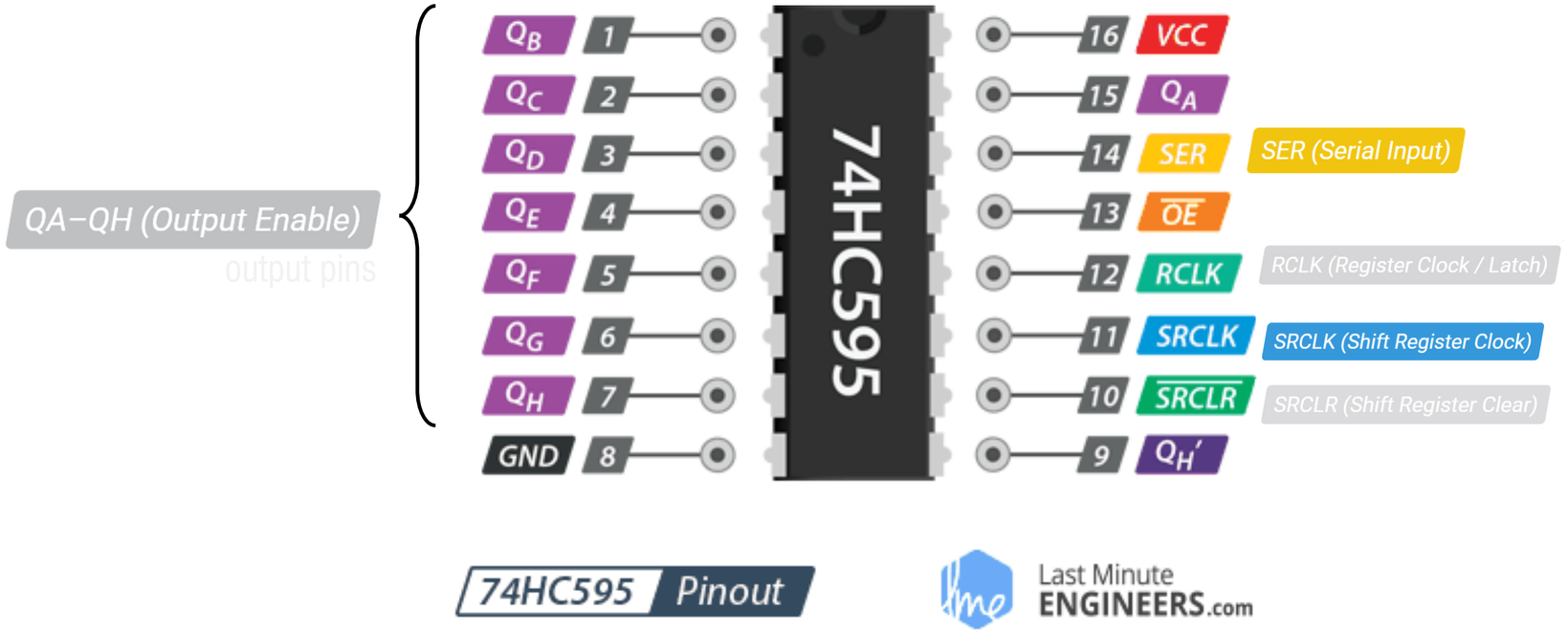
Shift Register - How – 74HC595



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74HC595 { 8bit Shift Register 0b00000100
8bit Storage Register

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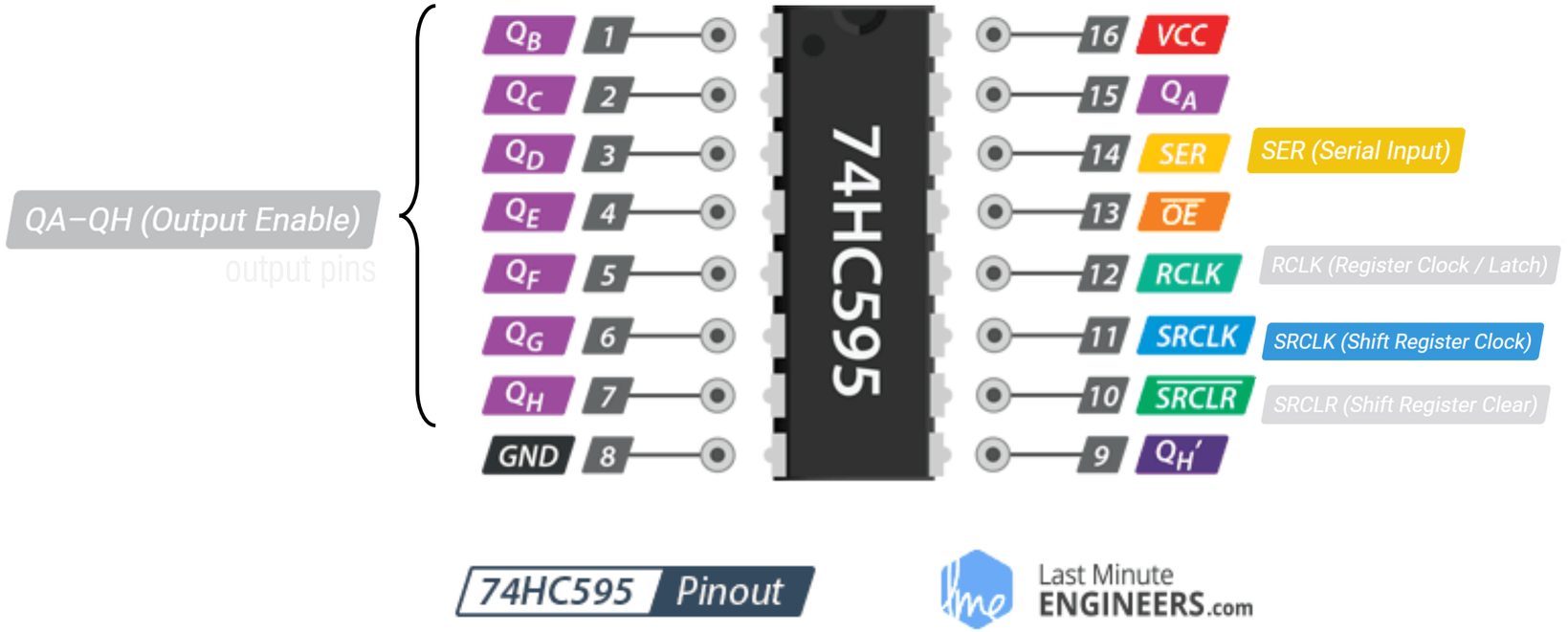
Shift Register - How – 74HC595



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74HC595 { 8bit Shift Register / 8bit Storage Register

0b00001000

Clock ticks 8 times

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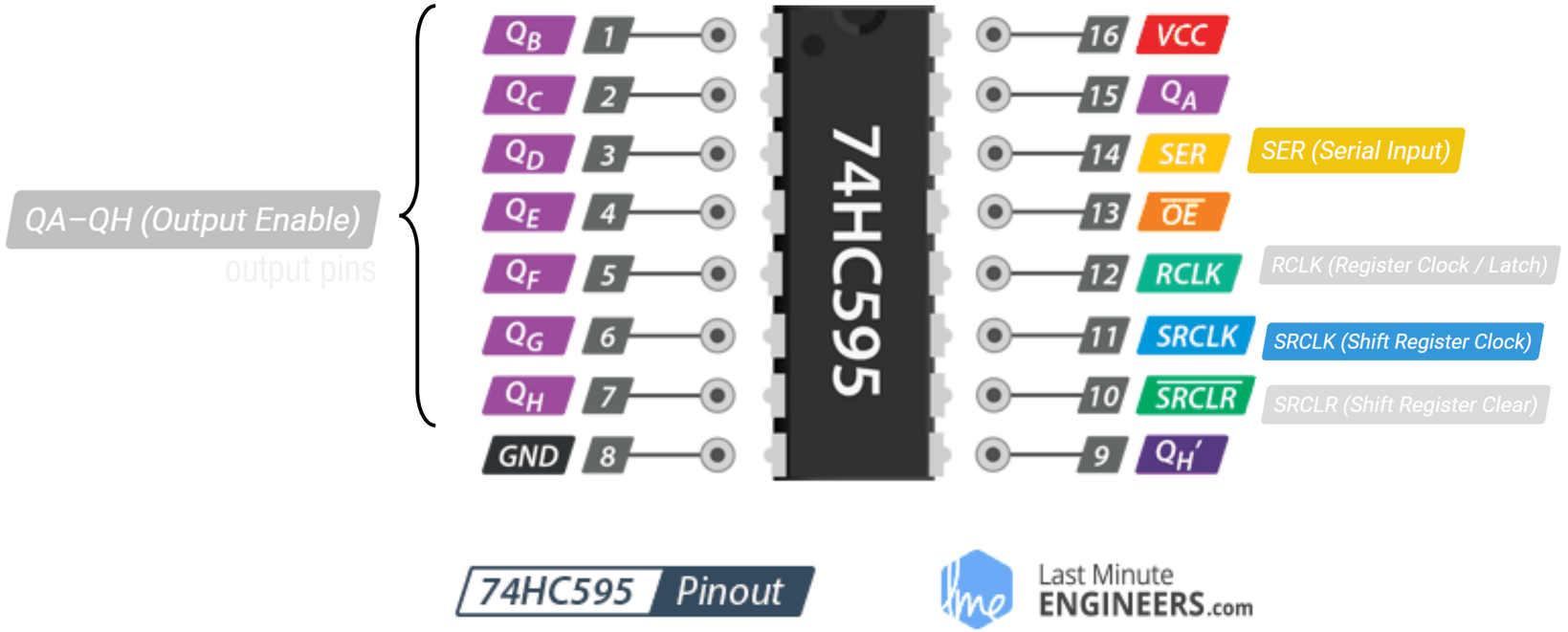
Shift Register - How – 74HC595



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74HC595 { 8bit Shift Register 0b10000000
8bit Storage Register

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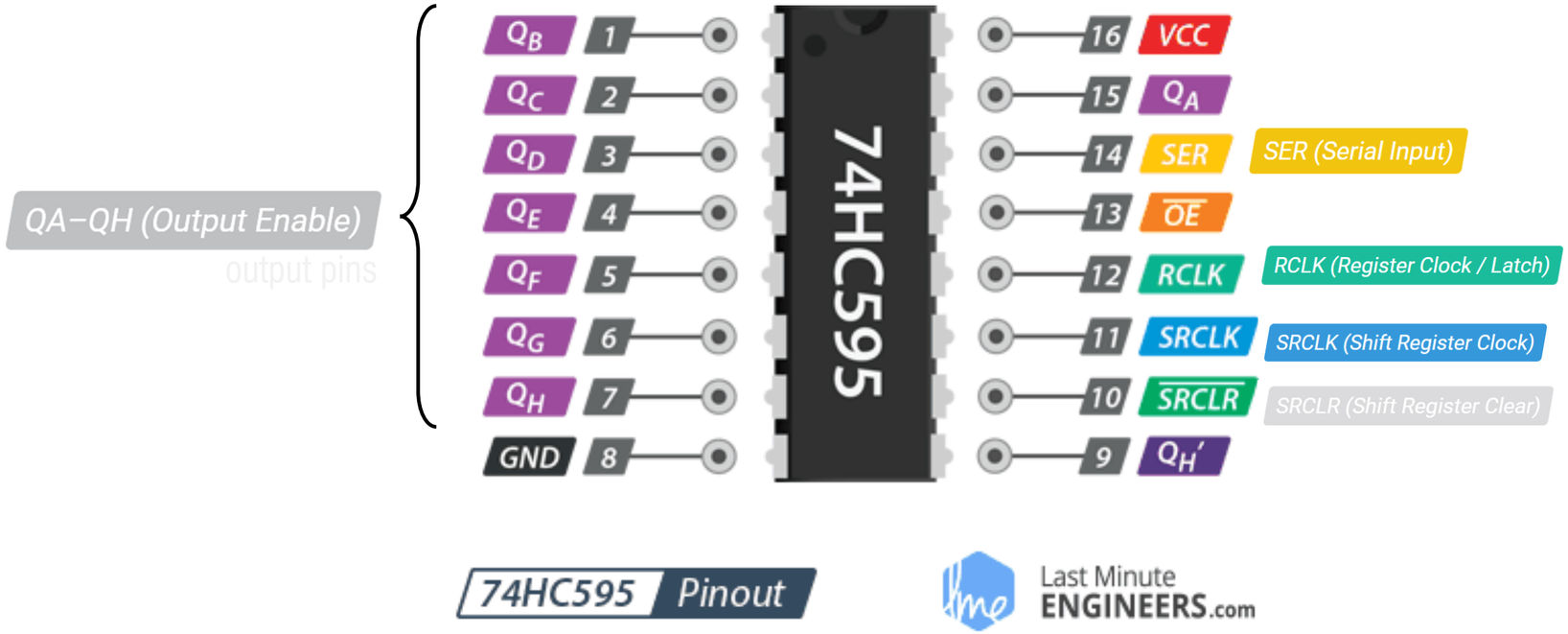
Shift Register - How – 74HC595



74HC595 Pinout

Last Minute ENGINEERS.com

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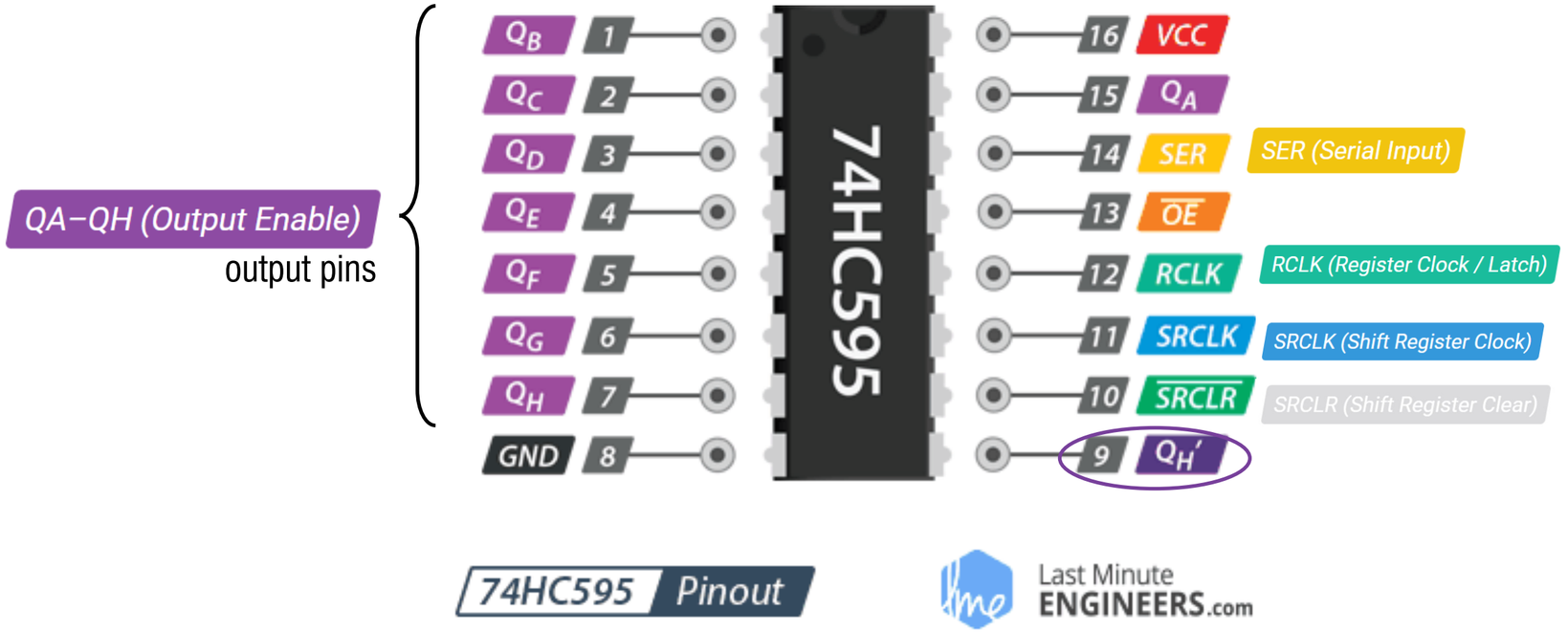
Shift Register - How – 74HC595



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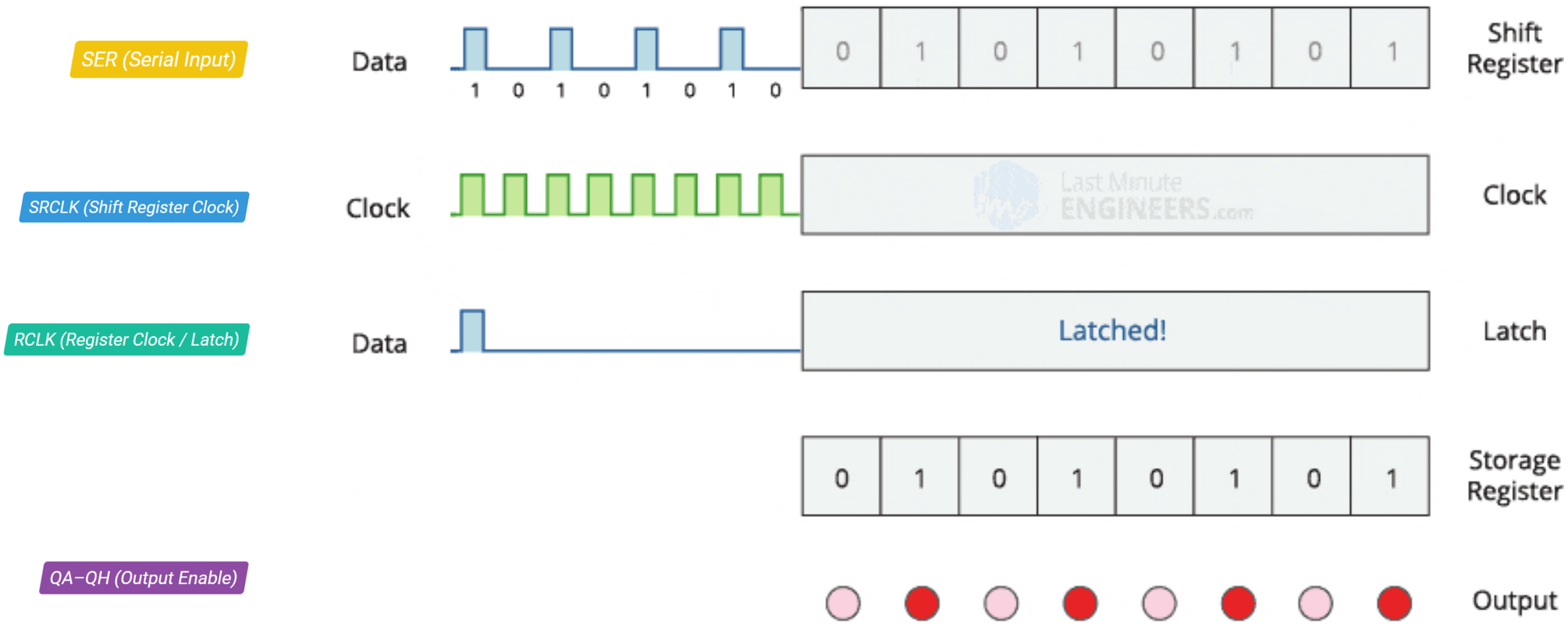

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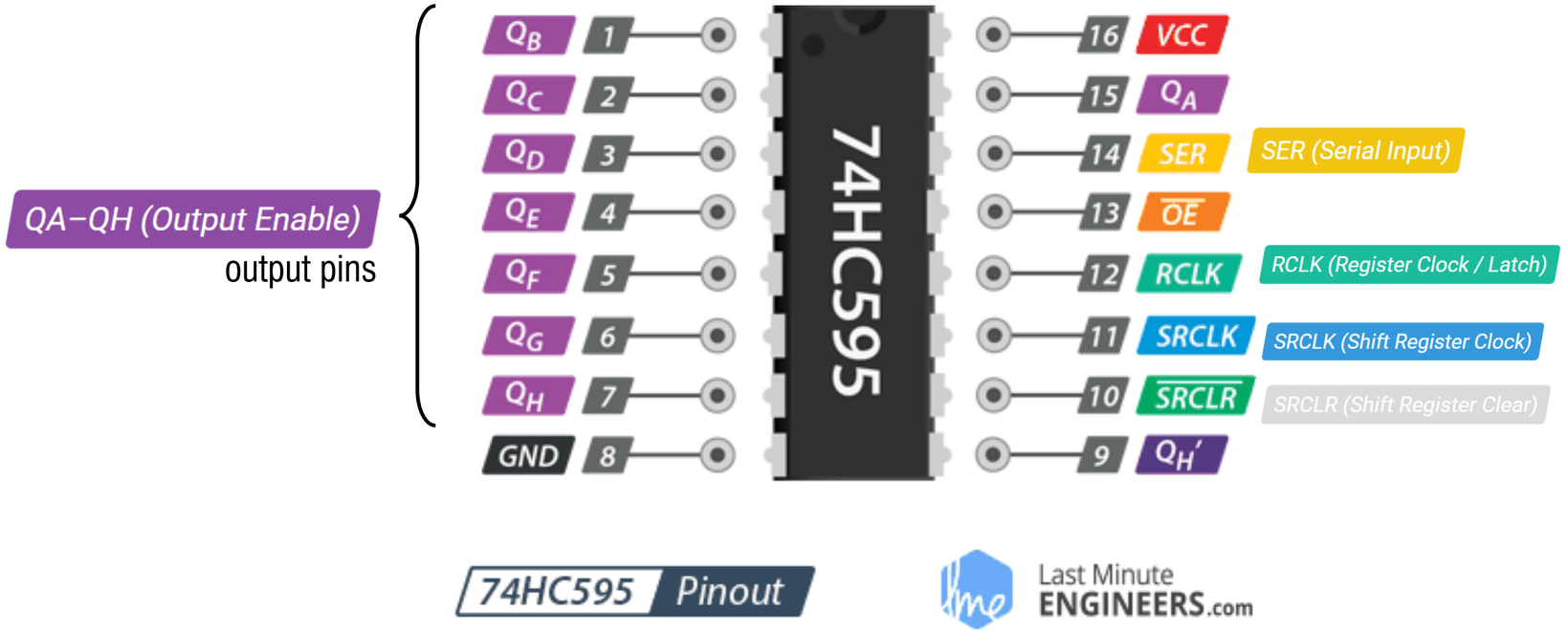
Shift Register - How – 74HC595



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- 8bit Storage Register 0b10000000

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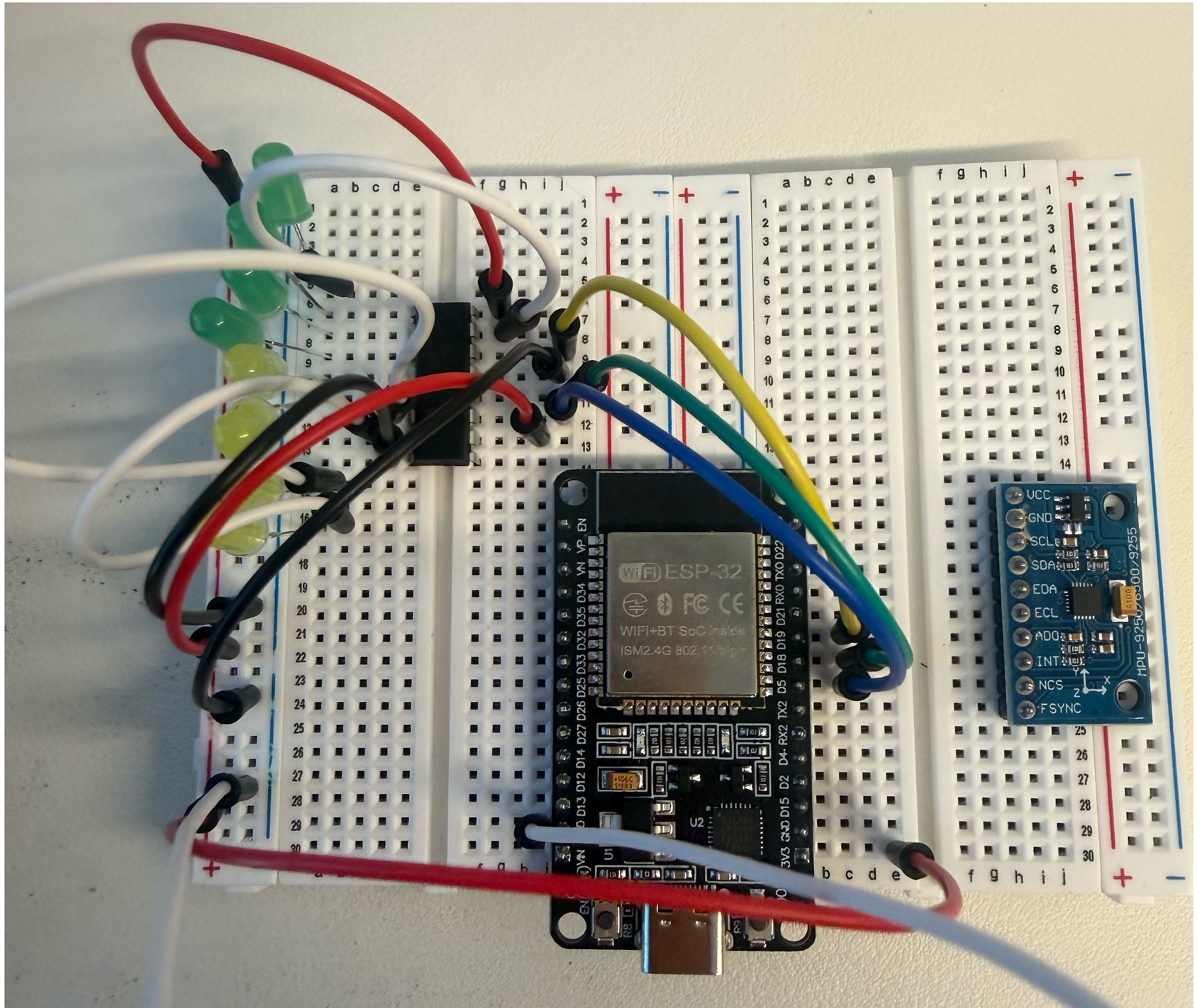
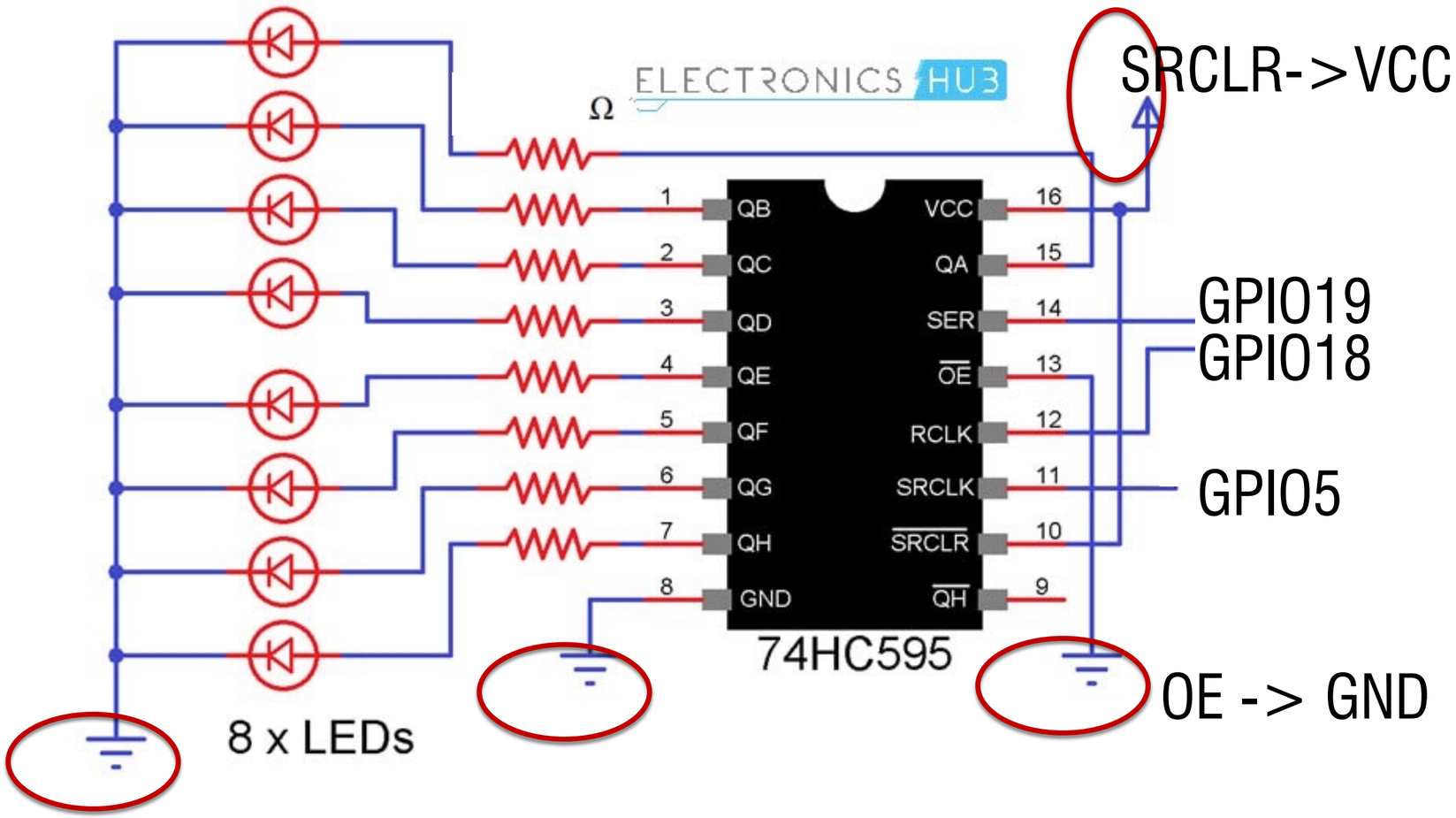
74HC595 - Wiring



74HC595 - Wiring



SER->Data
RCLK->Latch
SRCLK->Clock

0b10000000

74HC595 - Basic coding - light up one LED

```
int latchPin = 18;  // Latch pin of 74HC595
int clockPin = 5; // Clock pin of 74HC595
int dataPin = 19;  // Data pin of 74HC595

void setup()
{
 pinMode(latchPin, OUTPUT);
 pinMode(dataPin, OUTPUT);
 pinMode(clockPin, OUTPUT);
}

void loop()
{
 digitalWrite(dataPin, HIGH);
 digitalWrite(clockPin, HIGH);
 digitalWrite(clockPin, LOW);

 for(int i=0; i<7; i++)
 {
  digitalWrite(dataPin, LOW);
  digitalWrite(clockPin, HIGH);
  digitalWrite(clockPin, LOW);
 }

 digitalWrite(latchPin, HIGH);
 digitalWrite(latchPin, LOW);

 delay(4000);
}
```

0b10000000

Push one bit of ‘1’ into the shift register
Tick the clock pin

Shift seven bit of ‘0’ into the shift register

Tick the clock pin each time a new data is in

Set the latch pin to push data to the storage register that sent to output

bitSet()

[Bits and Bytes]

Description

Sets (writes a 1 to) a bit of a numeric variable.

Syntax

```
bitSet(x, n)
```

Parameters

`x`: the numeric variable whose bit to set.

`n`: which bit to set, starting at 0 for the least-significant (rightmost) bit.

Returns

`x`: the value of the numeric variable after the bit at position `n` is set.

```
byte myByte = 0b00001100; // Initial value: 12 in binary

bitSet(myByte, 2);
void setup() {
    Serial.begin(9600);
    Serial.println(myByte, BIN);
}
```

shiftOut()

Last revision • 04/29/2025

Description

Shifts out a byte of data one bit at a time. Starts from either the most (i.e. the leftmost) or least (rightmost) significant bit. Each bit is written in turn to a data pin, after which a clock pin is pulsed (taken high, then low) to indicate that the bit is available.

Note: If you're interfacing with a device that's clocked by rising edges, you'll need to make sure that the clock pin is low before the call to `shiftOut()`, e.g. with a call to `digitalWrite(clockPin, LOW)`.

This is a software implementation; see also the **SPI library**, which provides a hardware implementation that is faster but works only on specific pins.

Syntax

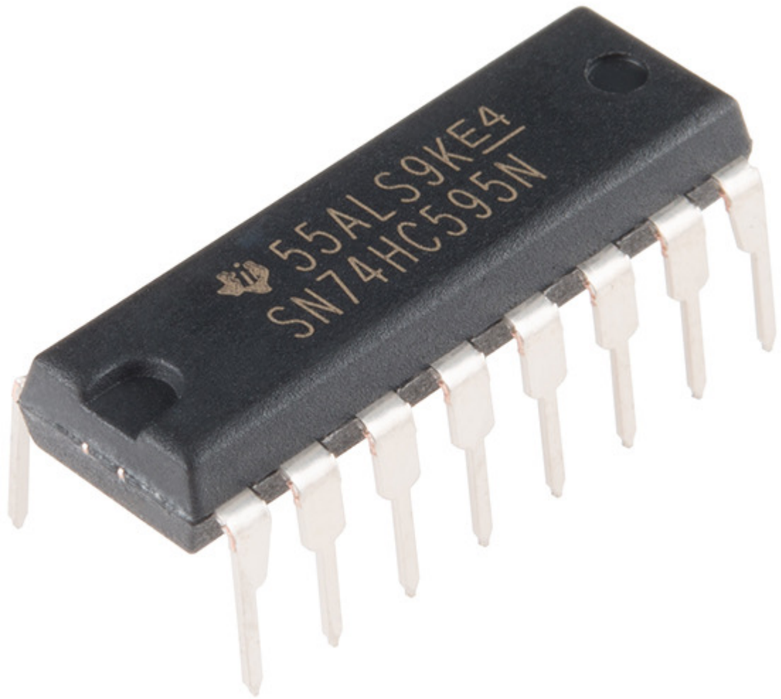
Use the following function to write out data synchronously through a pin:

```
shiftOut(dataPin, clockPin, bitOrder, value)
```

74HC595 - Controlling shift register with built-in functions

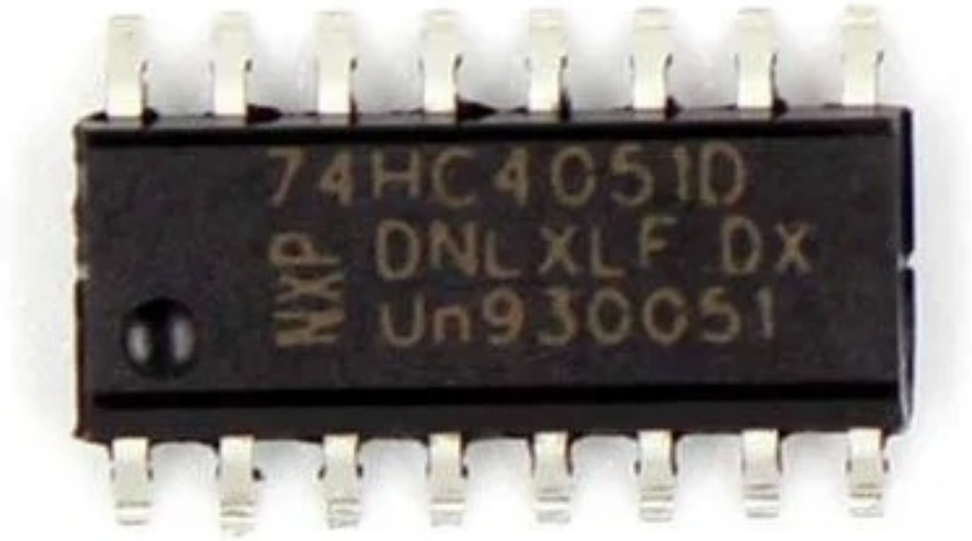
```
byte led_data = 0;                    // Variable to hold the pattern of which LEDs are currently turned on or off

void setup()
{
  pinMode(latchPin, OUTPUT);
  pinMode(dataPin, OUTPUT);
  pinMode(clockPin, OUTPUT);
}

void loop()
{
  led_data = 0;
  updateShiftRegister();
  delay(500);

  for(int i=0; i<8; i++)
  {
    bitSet(led_data, i);              // Set the bit that controls that LED in the variable 'led_data'
    updateShiftRegister();
    delay(500);
  }
}

void updateShiftRegister()
{
   digitalWrite(latchPin, LOW);
   shiftOut(dataPin, clockPin, LSBFIRST, led_data);   //Shifts out a byte of data one bit at a time
   digitalWrite(latchPin, HIGH);                      //putting the latch Pin HIGH
}
```

Shift Register Limitation?



Multiplexer

74HC4051



Multiplexer

Multiplexer (MUX): Directs 8 input channels into 1 output (e.g., reading 8 sensors using only 1 Analog-to-Digital converter pin).

Demultiplexer (DEMUX): Directs 1 input into 8 possible output channels.

Bidirectional: Unlike shift registers, signals can flow **both ways** because the internal connections are essentially low-resistance electronic switches.

A **Shift Register** "remembers" a state and holds it.
A **Multiplexer** is just a “bridge.” it just creates a path for a signal to travel through (no data storage).

Assignment – April 30 EOD